Procedure

1. Open Android Studio IDE → File → New → New Project → specify the application name "CRUD" and company domain "com.mad.crud" → click "next" → choose Minimum SDK "API 17:Android 4.2(Jelly Bean)" → click "Next" → choose "Blank Activity" → click "next" → specify the Activity Name "MainActivity" → click "Finish".
2. Open MainActivity.java under app/java/ crud.mad.com. crud and type the following codes:

MainActivity.java

```
package crud.mad.com.crud;

import android.support.v7.app.AppCompatActivity; // Non-AndroidX
import android.app.AlertDialog;
import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import android.view.View;


public class MainActivity extends AppCompatActivity {
   DatabaseHelper myDb;
   EditText editName, editDesig, editId;
   Button btnAdd, btnView, btnUpdate, btnDelete;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      myDb = new DatabaseHelper(this);

      editName = (EditText) findViewById(R.id.et_name);
      editDesig = (EditText) findViewById(R.id.et_desig);
      editId = (EditText) findViewById(R.id.et_id);
      btnAdd = (Button) findViewById(R.id.btn_add);
      btnView = (Button) findViewById(R.id.btn_view);
      btnUpdate = (Button) findViewById(R.id.btn_update);
      btnDelete = (Button) findViewById(R.id.btn_delete);

      // CREATE
      btnAdd.setOnClickListener(new View.OnClickListener() {
         @Override
         public void onClick(View v) {
            boolean inserted = myDb.insertEmployee(editName.getText().toString(),
editDesig.getText().toString());
```

```
            Toast.makeText(MainActivity.this, inserted ? "Success" : "Failed",
Toast.LENGTH_SHORT).show();
        }
    });

    // READ
    btnView.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            Cursor res = myDb.getAllEmployees();
            if(res.getCount() == 0) { showMsg("Error", "No records found"); return; }
            StringBuffer buffer = new StringBuffer();
            while (res.moveToNext()) {
                buffer.append("ID: "+res.getString(0)+"\nName: "+res.getString(1)+"\nPos:
"+res.getString(2)+"\n\n");
            }
            showMsg("Employees", buffer.toString());
        }
    });

    // UPDATE
    btnUpdate.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            boolean updated = myDb.updateEmployee(editId.getText().toString(),
editName.getText().toString(), editDesig.getText().toString());
            Toast.makeText(MainActivity.this, updated ? "Updated" : "Error",
Toast.LENGTH_SHORT).show();
        }
    });

    // DELETE
    btnDelete.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            Integer deleted = myDb.deleteEmployee(editId.getText().toString());
            Toast.makeText(MainActivity.this, deleted > 0 ? "Deleted" : "ID not found",
Toast.LENGTH_SHORT).show();
        }
    });
    Button btnClear = (Button) findViewById(R.id.btn_clear);
    btnClear.setOnClickListener(new View.OnClickListener() {
        public void onClick(View v) {
            // Clear all EditText fields
            editName.setText("");
            editDesig.setText("");
            editId.setText("");


            Toast.makeText(MainActivity.this, "Fields Cleared",
```

```
Toast.LENGTH_SHORT).show();
            }
        });
    }

    public void showMsg(String title, String msg) {
        AlertDialog.Builder builder = new AlertDialog.Builder(this);
        builder.setCancelable(true);
        builder.setTitle(title).setMessage(msg).show();
    }

}
```

3. Right click on crud.mad.com. crud package → New → Java Class → Name: DatabaseHelper → ok. Type the following codes in DatabaseHelper.java,

DatabaseHelper.java

```
package crud.mad.com.crud;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DatabaseHelper extends SQLiteOpenHelper {
    public static final String DATABASE_NAME = "Company.db";
    public static final String TABLE_NAME = "employee_table";
    public static final String COL_1 = "ID";
    public static final String COL_2 = "NAME";
    public static final String COL_3 = "DESIGNATION";

    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE " + TABLE_NAME + " (ID INTEGER PRIMARY
KEY AUTOINCREMENT, NAME TEXT, DESIGNATION TEXT)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
        onCreate(db);
    }

    public boolean insertEmployee(String name, String desig) {
        SQLiteDatabase db = this.getWritableDatabase();
```

```
        ContentValues contentValues = new ContentValues();
        contentValues.put(COL_2, name);
        contentValues.put(COL_3, desig);
        long result = db.insert(TABLE_NAME, null, contentValues);
        return result != -1;
    }

    public Cursor getAllEmployees() {
        SQLiteDatabase db = this.getWritableDatabase();
        return db.rawQuery("SELECT * FROM " + TABLE_NAME, null);
    }

    public boolean updateEmployee(String id, String name, String desig) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(COL_1, id);
        contentValues.put(COL_2, name);
        contentValues.put(COL_3, desig);
        db.update(TABLE_NAME, contentValues, "ID = ?", new String[]{id});
        return true;
    }

    public Integer deleteEmployee(String id) {
        SQLiteDatabase db = this.getWritableDatabase();
        return db.delete(TABLE_NAME, "ID = ?", new String[]{id});
    }
}
```

4. Open activity_main.xml under app/res/layout and type the following codes:

activity_main.xml

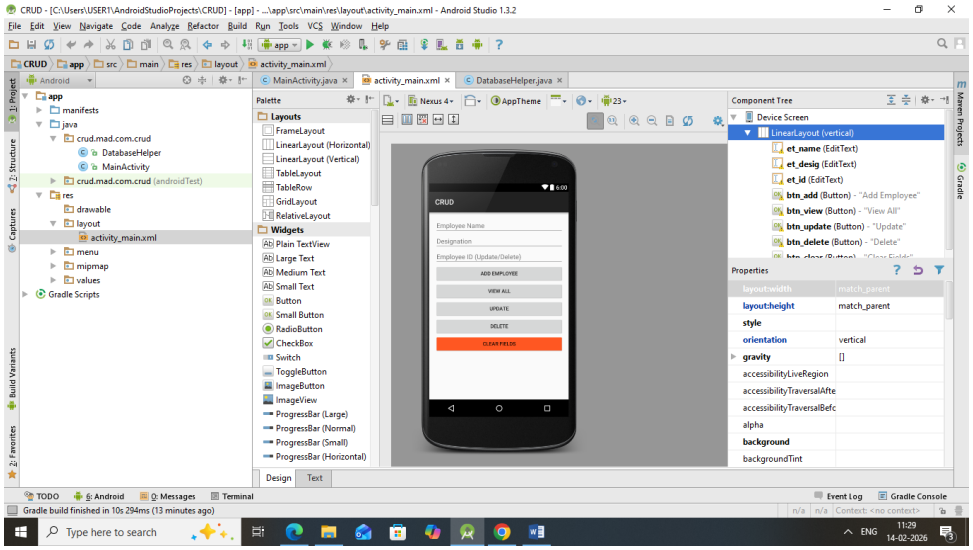
```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="16dp">

<EditText android:id="@+id/et_name" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Employee Name"/>
<EditText android:id="@+id/et_desig" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Designation"/>
<EditText android:id="@+id/et_id" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Employee ID (Update/Delete)"/>

<Button android:id="@+id/btn_add" android:layout_width="match_parent"
android:layout_height="wrap_content" android:text="Add Employee"/>
<Button android:id="@+id/btn_view" android:layout_width="match_parent"
android:layout_height="wrap_content" android:text="View All"/>
```

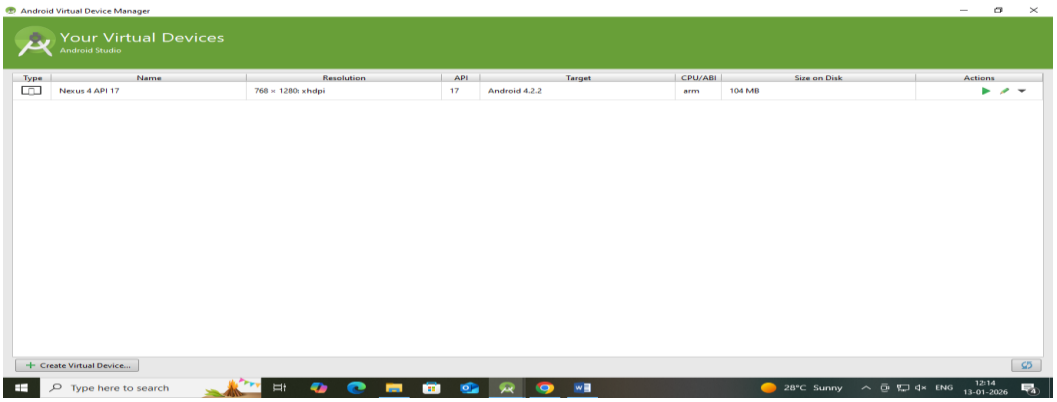
```
<Button android:id="@+id/btn_update" android:layout_width="match_parent"
android:layout_height="wrap_content" android:text="Update"/>
<Button android:id="@+id/btn_delete" android:layout_width="match_parent"
android:layout_height="wrap_content" android:text="Delete"/>
   <Button
      android:id="@+id/btn_clear"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Clear Fields"
      android:backgroundTint="#FF5722"/>
</LinearLayout>
```

5. The design of the application will be as follows:



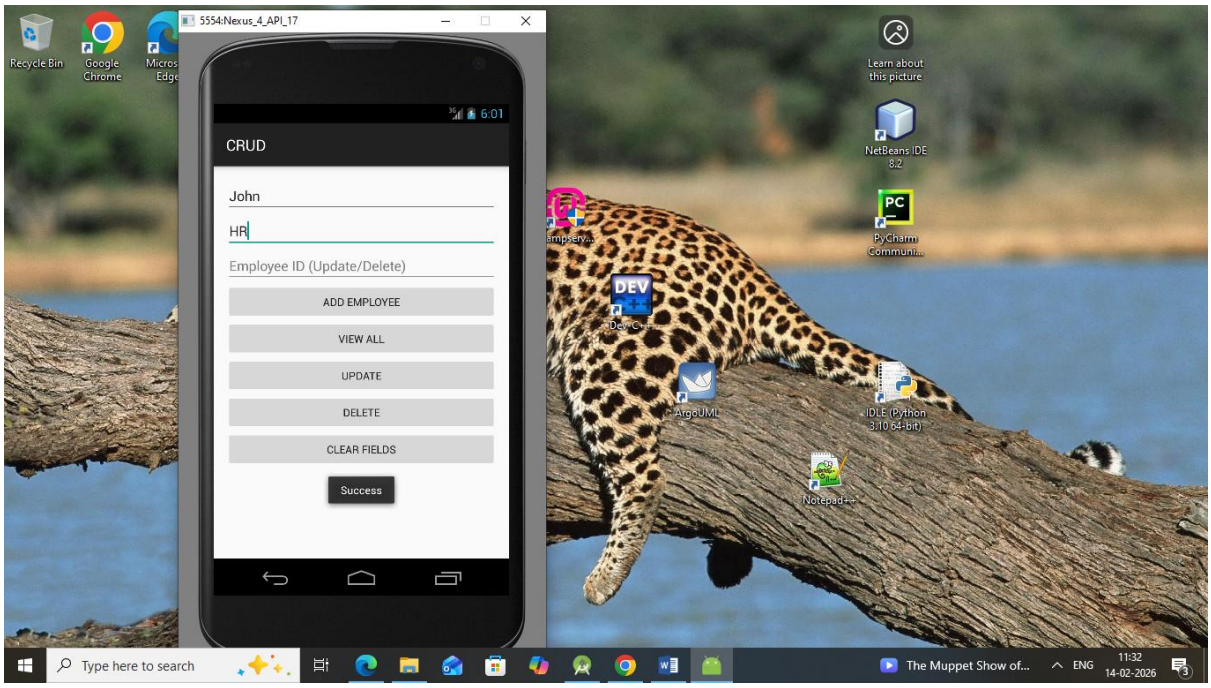
6. Go to Tools → android → AVD Manager → click "+ create a virtual device" → select "phone" from category → select "Nexus 4" from the list → click "next" → select Release name: Jelly Bean, API Level: 17, ABI: armeabi-v7a, Target: Android 4.2.2 from the list → click "next" → Choose orientation "portrait" → click "finish".

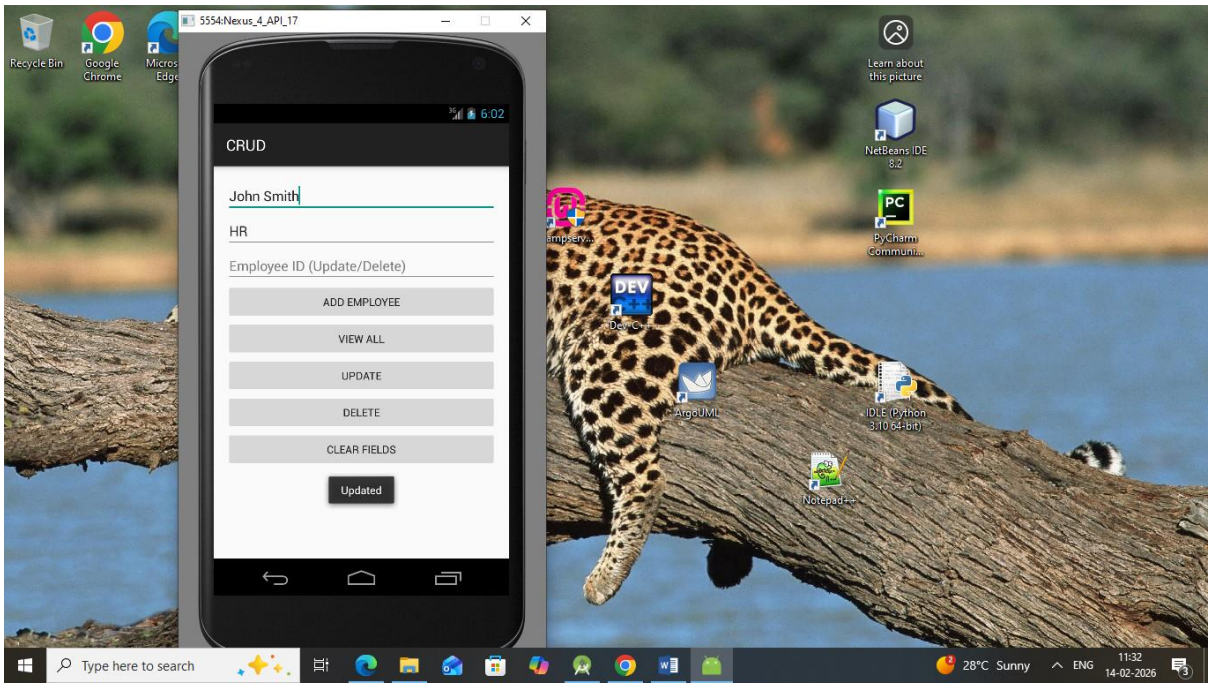
The following window will appear after configuring AVD:

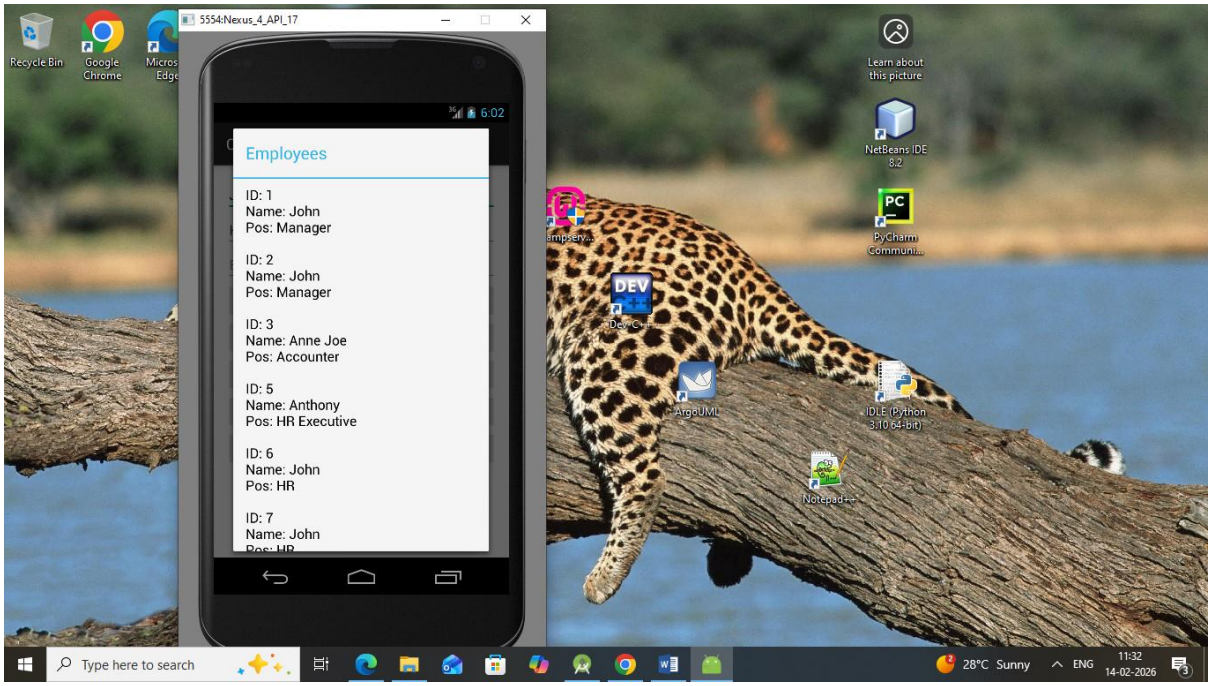


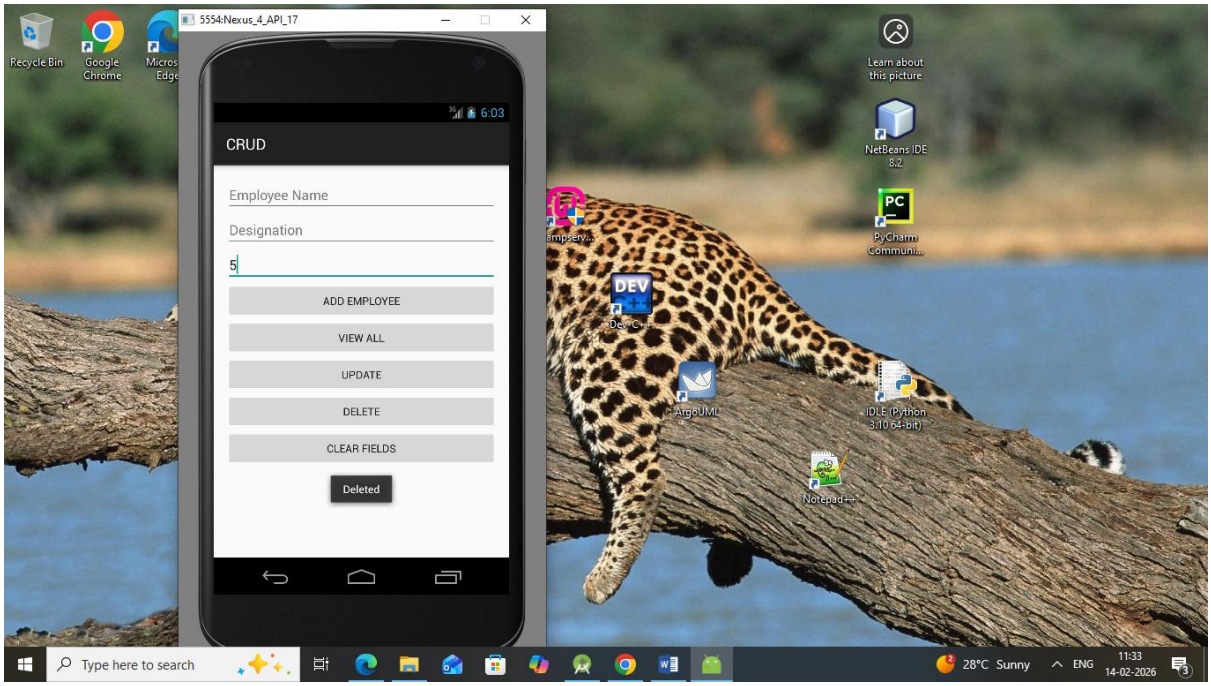
7. Click “Run app” button in the Android Studio → choose android virtual device → click “ok”.

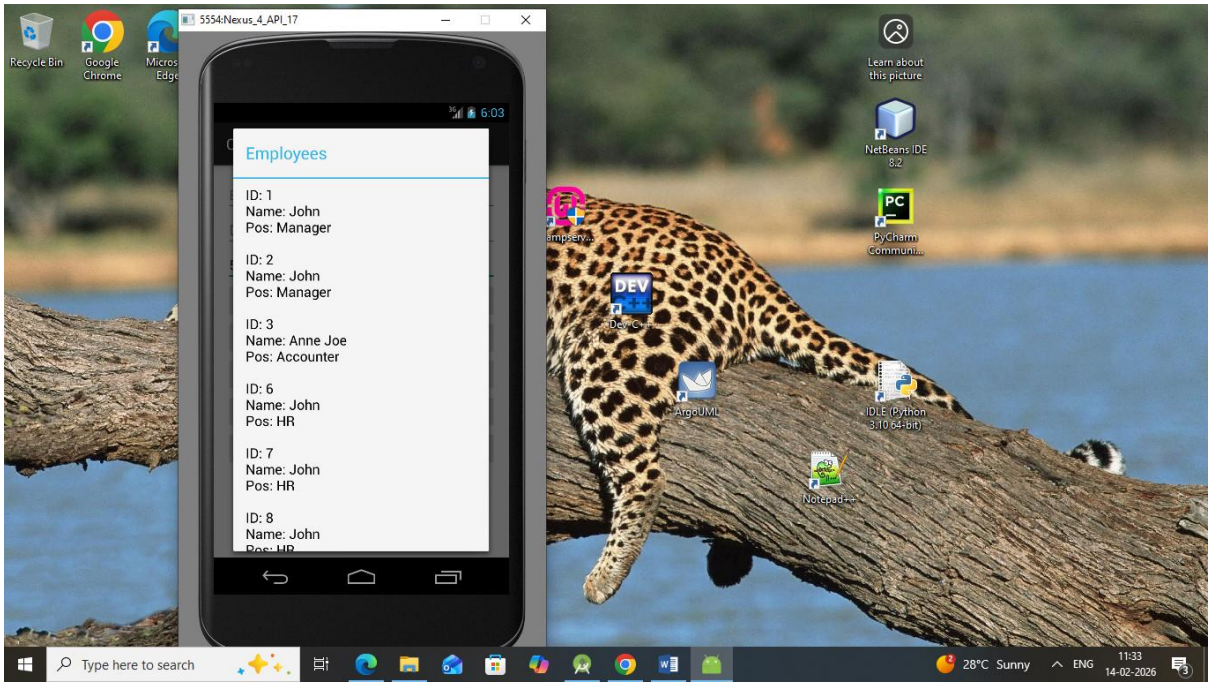
Output

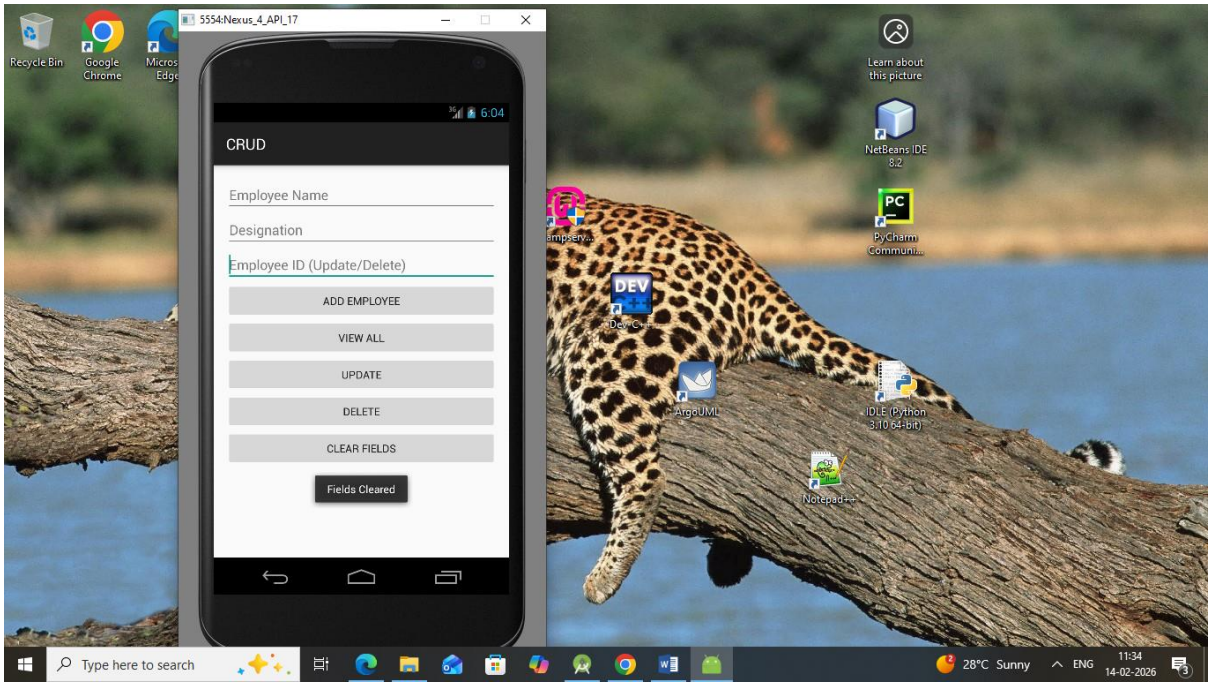





5554:Nexus_4_API_17
6:02
Employees
ID: 1
Name: John
Pos: Manager
ID: 2
Name: John
Pos: Manager
ID: 3
Name: Anne Joe
Pos: Accounter
ID: 5
Name: Anthony
Pos: HR Executive
ID: 6
Name: John
Pos: HR
ID: 7
Name: John


5554:Nexus_4_API_17
6:03
CRUD
Employee Name
Designation
5
ADD EMPLOYEE
VIEW ALL
UPDATE
DELETE
CLEAR FIELDS
Deleted


5554:Nexus_4_API_17
6:03
Employees
ID: 1
Name: John
Pos: Manager
ID: 2
Name: John
Pos: Manager
ID: 3
Name: Anne Joe
Pos: Accounter
ID: 6
Name: John
Pos: HR
ID: 7
Name: John
Pos: HR
ID: 8
Name: John


5554:Nexus_4_API_17
6:04
CRUD
Employee Name
Designation
Employee ID (Update/Delete)
ADD EMPLOYEE
VIEW ALL
UPDATE
DELETE
CLEAR FIELDS
Fields Cleared